

General-Relativistic Cyclic Cosmology from Holographic Saturation: An Effective Framework with Computed Substrate and Tensor Spectra

Paul Jarvis^{1,*}

¹*Independent Researcher, United Kingdom*

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We propose a cyclic cosmological framework in which the big-bang singularity is replaced by an information-saturation event that triggers a transition from gravitational collapse to expansion, modelled within general relativity as an Israel–Darmois junction across a spacelike hypersurface at a critical density. This paper’s results fall into two classes, which we keep explicitly separate. Three results are computed outputs of a single, publicly available numerical pipeline. First, the integrated spectral weight ρ_0 of the mutual-information graph Laplacian at the hyper-stiff point ($\alpha = 6$) converges to a lattice-independent continuum value ($\rho_0 \approx 4.68$, varying by $\lesssim 4\%$ across $L = 4$ –16 toroidal lattices), and we show that this Weyl spectral dimension is a distinct quantity from the fluid equation-of-state dimension, with the fit prescription stated explicitly. Second, direct Mukhanov–Sasaki integration through the transient confinement spike $\rho_{\text{conf}}(a) = \rho_0 \chi(1 - \chi)^8$ on a radiation background, with adiabatic vacuum initial conditions and Bogoliubov-coefficient extraction verified by Wronskian conservation, yields a clean ultraviolet power law $|\beta_k|^2 \propto k^{-4}$ - a red tensor tilt under the standard convention - with the tilt insensitive to the spike duration δ while the amplitude varies by four orders of magnitude over $\delta \in [0.4, 0.6]$; this supersedes an earlier, uncomputed blue-tilt estimate and implies no PTA-band gravitational-wave signal from the mechanism as formulated [14]. Third, Heisenberg spin-chain benchmarks show that the measured and Luttinger-liquid-predicted mutual-information decay exponents correlate at $r \approx 0.96$ across the accessible anisotropy range, the strongest quantitative support for the underlying emergent-geometry construction. Two results constrain the framework and are reported as such rather than as resolved. The tanh envelope of the confinement profile drives the effective equation of state to $w \approx 2.1 > 1$, in tension with the causal bound $c_s \leq c$ under the adiabatic identification $c_s^2 = w$; admissible profiles must therefore either be redesigned to fall no faster than a^{-6} or be realised non-adiabatically. Separately, the exponential damping ansatz that drives cycle-to-cycle homogenisation converges generically for any positive damping coefficient, but a first-principles derivation of that coefficient, of the confinement potential’s sign reversal, and of a coarse-graining mechanism that demonstrably isotropises the background all remain open problems.

Keywords: Cyclic Cosmology, Holographic Principle, Quantum Entanglement, Bouncing Universes, Primordial Gravitational Waves

I. INTRODUCTION

Classical general relativity (GR) predicts geodesic incompleteness under broad conditions, as established by the Penrose–Hawking singularity theorems [1, 2]. This motivates the search for frameworks in which the big-bang singularity is replaced by a well-defined transition. Existing approaches include loop quantum cosmology (LQC) [3], conformal cyclic cosmology (CCC) [4], ekpyrotic and non-singular bounce models [5–7], and holographic or fuzzball-inspired constructions [8–10]. A recurring difficulty across the non-singular bounce literature is maintaining a causally and perturbatively consistent effective description through the bounce itself [7]; we encounter the same difficulty below (Sec. IX B) and report it as an open constraint rather than as resolved.

We develop a cyclic cosmology in which the singularity is replaced by an information-saturation event. The transition from collapse to expansion is implemented as a GR-consistent junction across a spacelike hypersurface at a

critical density. The underlying intuition is holographic: as a collapsing region approaches its saturation radius, its information content approaches the Bekenstein–Hawking bound, and the effective confinement potential undergoes a sign reversal.

Two methodological commitments distinguish the present version of this work from earlier drafts. First, we separate explicitly the components of the framework that are *derived* (spectral properties of the substrate, the computed tensor spectrum) from those that are *phenomenological ansätze* (the confinement potential, the saturation envelope, the cyclic damping law). Second, every quantitative claim in this paper is the output of a single, publicly available numerical pipeline (Sec. XIII); no figure or headline number is asserted independently of that pipeline. Where the honest computation contradicts earlier estimates - most notably for the tensor spectral tilt - the computed result supersedes the earlier claim, and we say so explicitly.

* mrpaulwjarvis@gmail.com

II. PRE-GEOMETRIC SUBSTRATE AND EMERGENT GEOMETRY

A. Substrate Assumptions

We assume that smooth geometry emerges natively from the quantum entanglement structure of an underlying pre-geometric many-body substrate [11–13]. The mutual information between two arbitrary sub-regions i and j ,

$$I(i, j) = S(\rho_i) + S(\rho_j) - S(\rho_{ij}), \quad (1)$$

serves as a relational metric proxy that maps spatial proximity. This mutual information matrix dictates a symmetric connectivity matrix A_{ij} , from which we construct the combinatorial graph Laplacian

$$L_{ij}^{\text{MI}} = \delta_{ij} \sum_k I_{ik} - I_{ij}. \quad (2)$$

No specific microscopic Hamiltonian or prior spatial embedding is assumed. All macroscopic phenomenological parameters are treated as coarse-grained descriptions whose exact microphysical derivations are deferred to future work.

B. Heisenberg-Chain Benchmarks

To test this construction against explicit quantum states, we track the low-lying spectrum of the graph Laplacian derived from ground states of the antiferromagnetic Heisenberg (XXZ) spin chain [15],

$$H = J \sum_{i=1}^{N-1} \vec{S}_i \cdot \vec{S}_{i+1}, \quad J > 0, \quad (3)$$

for chain lengths $N = 6\text{--}16$, using exact diagonalisation and mutual information computed from partial traces of the exact ground state. The full pipeline, including AICc model selection, leave-one-out sensitivity, and normalisation robustness checks, is implemented in `code/testing.py`, supplied with this paper [15].

The strongest quantitative support for the construction is the correspondence between the measured mutual-information decay exponent and the Luttinger-liquid prediction $\alpha_{\text{MI}} = -2\min(2K, 1/2K)$ across the anisotropy range $\Delta \in (-0.9, 1]$: the measured and predicted exponents correlate at $r \approx 0.96$, with near-exact agreement at the Heisenberg point ($\Delta = 1$: predicted -2.000 , measured -1.96). The low-lying spectrum is consistent with a smooth one-dimensional emergent geometry; however, we note two honest caveats established by the robustness suite in `code/testing.py`. First, information-criterion (AICc) model selection between linear and quadratic dispersion remains ambiguous at accessible system sizes, so we describe the dispersion as *consistent with* quadratic

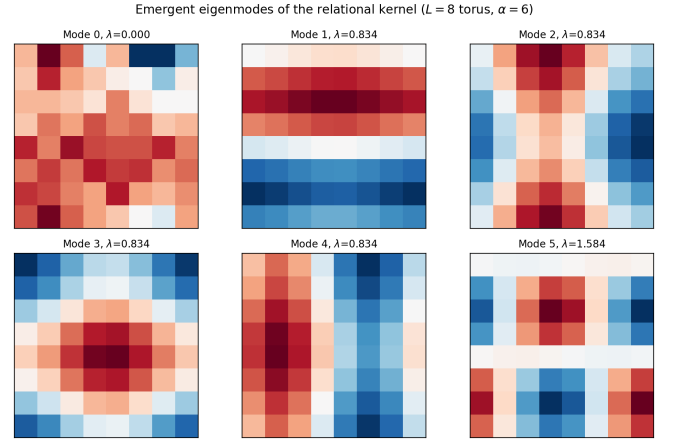


FIG. 1. Emergent eigenmodes of the relational kernel on the 8×8 torus at $\alpha = 6$. Mode 0 ($\lambda = 0.000$) is the invariant global ground state. Modes 1–4 form a four-fold degenerate tier at $\lambda = 0.834$ whose span provides two orthogonal spatial coordinate gradients over an isotropic 2D manifold. Higher modes are localised geometric harmonics. All eigenvalues shown are computed outputs of the pipeline of Sec. XIII.

rather than established. Second, the finite-size scaling of the spectral gap is well fit by $\lambda_1 = a/N^2 + b$ with a nonzero intercept b at accessible sizes; leave-one-out and normalisation-sensitivity checks indicate that the significance of b is not yet settled, and the pure scaling law $\lambda_1 \propto N^{-2}$ should be regarded as the large- N hypothesis rather than a demonstrated result.

C. Emergent Coordinate Charts on the 2D Torus

Numerical diagonalisation of a relational kernel $I_{ij} = 1/d_{ij}^\alpha$ on a discrete 2D toroidal lattice demonstrates spontaneous coordinate chart generation. As shown in Fig. 1 for the 8×8 torus at the hyper-stiff point $\alpha = 6$, mode 0 ($\lambda = 0$) is the flat global ground state, and the first excited tier is a *four-fold degenerate* multiplet at $\lambda = 0.834$ whose span contains two mutually orthogonal long-wavelength gradients - the coordinate charts of an emergent isotropic 2D manifold. (An earlier draft described this tier as a degenerate pair; the four-fold degeneracy is the correct statement for the torus, reflecting its $\pm k_x, \pm k_y$ symmetry.) Higher modes define localised geometric harmonics with a UV cutoff set by the lattice spacing.

III. SPECTRAL FOUNDATIONS OF THE SUBSTRATE ENERGY DENSITY

A. Canonical Partition Function and ρ_0

The effective macroscopic energy density of the substrate is constructed from the statistical mechanics of the

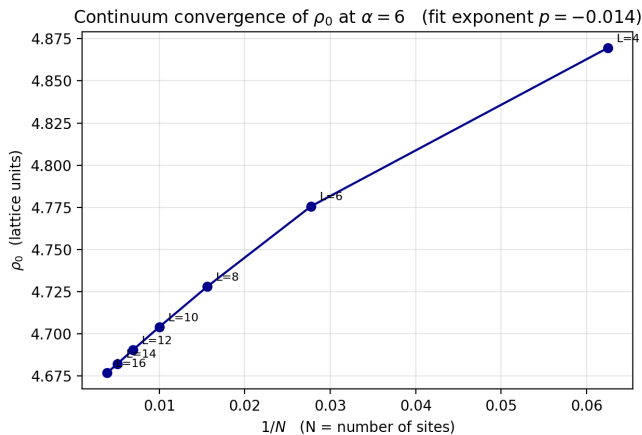


FIG. 2. Continuum convergence of ρ_0 at $\alpha = 6$ across toroidal lattices $L = 4$ – 16 . The fitted size exponent is $p = -0.014$, consistent with a lattice-independent continuum value $\rho_0 \approx 4.68$.

MI-Laplacian spectrum. Treating the eigenvalues $\{\lambda_n\}$ as the fundamental spectrum of structural fluctuations, we define the canonical partition function over excited modes ($\lambda_n > 0$) as

$$Z = \sum_{n=1}^{N-1} e^{-\beta \lambda_n}, \quad (4)$$

where β is an inverse entanglement temperature setting the coarse-graining scale. The internal energy density is

$$\rho_{\text{eff}} = -\frac{\partial \ln Z}{\partial \beta} = \frac{\sum_n \lambda_n e^{-\beta \lambda_n}}{\sum_n e^{-\beta \lambda_n}}, \quad (5)$$

and the integrated spectral weight at the hyper-stiff point is

$$\rho_0 = \int_{\lambda_{\min}}^{\lambda_{\max}} \lambda g(\lambda) d\lambda, \quad (6)$$

where $g(\lambda)$ is the MI-Laplacian spectral density.

Earlier drafts described ρ_0 as a “spectral invariant” on the basis of a single 8×8 evaluation. Here we establish the correct, weaker, and well-supported statement: ρ_0 converges to a lattice-independent continuum value. Scanning toroidal lattices from $L = 4$ to $L = 16$ at $\alpha = 6$ (Fig. 2) yields $\rho_0 = 4.870, 4.776, 4.728, 4.704, 4.691, 4.682, 4.677$, a monotone approach with total spread below 4% and a fitted size dependence $\rho_0 \propto N^p$ with $p = -0.014$, i.e. consistent with N -independence. We quote the continuum estimate $\rho_0 \approx 4.68$ (lattice units); the previously reported 8×8 value of 4.728 lies within 1.1% of it.

B. Physical Amplitude: The Roles of ε_0 and δ

Converting ρ_0 to physical units requires a microscopic energy scale ε_0 (energy per eigenvalue unit) and an effec-

tive lattice volume V :

$$\rho_{\text{phys}} = \frac{\rho_0 N \varepsilon_0}{V}. \quad (7)$$

Earlier drafts identified ε_0 as the sole free parameter of the amplitude sector. The computed tensor spectra of Sec. X show that this is not the case: at fixed ε_0 , the particle-production amplitude varies by four orders of magnitude as the spike duration δ ranges over $[0.4, 0.6]$, while the spectral *tilt* is insensitive to δ . The amplitude sector therefore carries (at least) two free parameters, ε_0 and δ , and no observational amplitude claim is made in this paper. The tilt and the ultraviolet power-law form of the spectrum are the parameter-robust outputs.

C. Two Distinct Dimension Concepts

A source of potential confusion in the literature on emergent geometry is the conflation of two distinct notions of dimension. We distinguish them here and state the fit prescription explicitly.

The *Weyl spectral dimension* $d_{\text{eff}}^{\text{Weyl}}$ is extracted from the MI-Laplacian eigenvalue counting function via

$$\lambda_n \propto n^{2/d_{\text{eff}}^{\text{Weyl}}}, \quad (8)$$

fitting the slope of $\log \lambda_n$ versus $\log n$ over the low-lying spectrum. All values in Table I use the fit window $n = 1$ – 12 on the 6×6 torus; the extracted dimension depends appreciably on this window, which earlier drafts left unstated. Appendix A 2 quantifies this dependence at $\alpha = 6$: $d_{\text{eff}}^{\text{Weyl}}$ drifts from 5.88 at $n_{\text{fit}} = 6$ down to 3.46 over the full available spectrum, and is not yet asymptotic at $n_{\text{fit}} = 12$. The quoted values should therefore be read as finite-window estimates rather than converged asymptotic dimensions.

The *fluid dimension* d_{fluid} is inferred from the equation-of-state scaling

$$\rho_{\text{eff}}(a) \propto a^{-3(1+w_{\text{eff}})}, \quad d_{\text{fluid}} = 3(1+w_{\text{eff}}) = \alpha, \quad (9)$$

where the second equality follows directly from $w_{\text{eff}} = \alpha/3 - 1$.

TABLE I. Weyl spectral dimension $d_{\text{eff}}^{\text{Weyl}}$ versus fluid dimension $d_{\text{fluid}} = \alpha$ at selected values of the MI decay exponent α (6×6 torus, fit window $n = 1$ – 12). All values are outputs of the pipeline of Sec. XIII.

α	w_{eff}	$d_{\text{eff}}^{\text{Weyl}}$	d_{fluid}
1.5	−0.500	11.14	1.5
2.5	−0.167	6.68	2.5
3.5	+0.167	5.01	3.5
4.5	+0.500	4.26	4.5
6.0	+1.000	3.81	6.0
7.0	+1.333	3.70	7.0

These two quantities are *not* equal in general. At the hyper-stiff transition ($\alpha = 6$, $w_{\text{eff}} = 1$) the Weyl spectral dimension is $d_{\text{eff}}^{\text{Weyl}} = 3.81$, not 8. The value $d_{\text{eff}} \approx 8$ at $\alpha = 1.5$ ($w_{\text{eff}} = -0.5$) belongs to a distinct regime from the hyper-stiff transition. These two regimes must not be conflated.

IV. HOLOGRAPHIC SATURATION AND CONFINEMENT POTENTIAL

A. Information Bound

A collapsing region of radius r has an information storage capacity strictly bounded by its boundary area according to the holographic principle and its covariant generalisation [8–10]:

$$I_{\text{max}}(r) = \frac{A}{4} = \pi r^2, \quad (10)$$

reaching complete information saturation at a limiting minimal radius $r = r_0$.

B. Sign Reversal of V_{conf} (Ansatz)

We introduce an effective macroscopic confinement potential $V_{\text{conf}}(\rho)$:

$$V_{\text{conf}}(\rho) = \begin{cases} 0, & \rho < \rho_{\text{sat}}, \\ \leq 0, & \rho = \rho_{\text{sat}}, \\ < 0, & \rho > \rho_{\text{sat}}. \end{cases} \quad (11)$$

We emphasise that Eq. (11) is a *phenomenological ansatz*: the sign flip is postulated as the macroscopic signature of a conjectured microphysical phase transition in which the attractive gravitational degrees of freedom condense and manifest net repulsion at extreme densities. No microscopic Hamiltonian realising this transition is provided here, and its construction is the principal open problem of the framework.

V. GR-CONSISTENT JUNCTION FORMULATION

A. Causal Boundary

The collapse–expansion bounce is modelled entirely within classical GR across a spacelike hypersurface Σ , which behaves as a strict localised causal boundary.

B. Israel–Darmois Junction

Matching the interior collapsing and exterior expanding spacetimes requires satisfying the Israel–Darmois

junction conditions. The induced metric h_{ab} must be continuous:

$$[h_{ab}] = h_{ab}^+ - h_{ab}^- = 0. \quad (12)$$

The jump in extrinsic curvature K_{ab} is related to the surface stress-energy S_{ab} :

$$[K_{ab}] - h_{ab}[K] = -8\pi G S_{ab}. \quad (13)$$

We *assume* that at global holographic saturation $S_{ab} \rightarrow 0$, so that the bounce is driven by a geometric momentum reflection across Σ rather than exotic matter fields. This assumption is physically motivated by the saturation condition but is not derived from it here; establishing $S_{ab} \rightarrow 0$ from the microphysics is an open task.

C. Modified Friedmann Equation (Ansatz)

Near the saturation hypersurface the effective dynamics are taken to be

$$H^2 = \frac{\rho}{3} [1 - \chi(\rho)], \quad (14)$$

with the smooth saturation envelope

$$\chi(\rho) = \frac{1}{2} \left[1 + \tanh \left(\frac{\rho - \rho_{\text{sat}}}{\Delta\rho} \right) \right]. \quad (15)$$

The tanh form is chosen for smoothness and is not derived; alternative envelopes with the same limits are equally admissible at this stage, and Sec. IX B shows that the choice is in fact *constrained* by causality.

VI. ENTROPY INHERITANCE AND RENORMALISATION

In a cyclic universe, the absolute entropy produced during the expansion phase of cycle n , combined with the residual entropy inherited from cycle $n - 1$, dictates the total entropy dumped into the subsequent bounce throat:

$$S_{\text{dump}}^{(n)} = S_{\text{prod}}^{(n)} + S_{\text{inh}}^{(n)}. \quad (16)$$

The dimensionless renormalised inhomogeneity ratio is

$$\tilde{S}^{(n+1)} = \frac{S_{\text{inh}}^{(n+1)}}{S_{\text{max}}^{(n+1)}}. \quad (17)$$

If the total available phase space S_{max} expands at a faster geometric rate than the inherited entropy across each bounce, the universe re-enters the expansion phase in an effectively low-entropy configuration without non-unitary entropy destruction. Whether the substrate dynamics enforce this hierarchy of growth rates is not derived here; it is the defining assumption of the cyclic sector.

VII. HOMOGENISATION (CONJECTURE)

The generation of an isotropic background from an initially asymmetrical collapse is conjectured to proceed in two stages: chaotic mixing during the late contracting phase, followed by holographic coarse-graining of the remaining bulk fluctuations into $\mathcal{O}(A/\ell_P^2)$ boundary degrees of freedom near saturation.

We flag this mechanism as a conjecture rather than a result. In direct numerical tests, a naive real-space block decimation (Kron reduction) of anisotropic MI graphs does *not* by itself reduce the anisotropy of the effective couplings; the anisotropy measure fluctuates rather than flowing to zero under successive decimation steps. If holographic coarse-graining suppresses anisotropy, it must do so through a different operation (for example, spectral projection onto low-lying modes), and demonstrating this explicitly is required before the mechanism can be claimed. Consequently, no solution to the flatness or horizon problems is claimed in this paper.

VIII. CYCLIC ATTRACTOR (PHENOMENOLOGICAL)

Each discrete cosmic bounce defines a discrete map $\mathcal{R}$ on the space of effective background couplings:

$$\mathcal{R} : \{\lambda^{(n)}\} \mapsto \{\lambda^{(n+1)}\}. \quad (18)$$

We model the cycle-to-cycle dilution of inhomogeneity by the exponential damping ansatz

$$\mathcal{D}_n = \exp\left(-\frac{cN}{\rho_{\text{throat}}^{(n)}}\right), \quad 0 < \mathcal{D}_n < 1, \quad (19)$$

where $\rho_{\text{throat}}^{(n)}$ is the partition-function energy density of the MI-Laplacian evaluated at the n -th throat and c is a dimensionless coefficient.

Two honest statements replace the claims of earlier drafts. First, convergence of $\tilde{S}(n)$ to zero under Eq. (19) is *generic*, not fine-tuned: because $0 < \mathcal{D}_n < 1$ for any $c > 0$ while S_{max} grows geometrically, the renormalised ratio decays for essentially any coefficient (Fig. 3 shows $c = 0.05, 0.22$, and 1.0). The existence of the fixed point $\tilde{S}^* = 0$ is therefore a structural property of the ansatz, and Fig. 3 is an illustration of that ansatz, not a derivation. Second, the coefficient c is a free parameter: direct attempts to derive it from the Gibbs entropy of the MI-Laplacian partition function do not yield a stable value, so we do not attach significance to any particular choice. A first-principles derivation of the damping rate is an open problem.

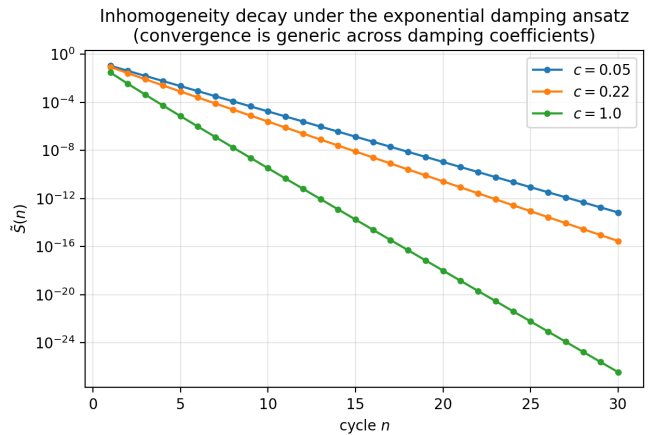


FIG. 3. Decay of the renormalised inhomogeneity ratio $\tilde{S}(n)$ under the exponential damping ansatz of Eq. (19) for three values of the damping coefficient c , with $\rho_{\text{throat}}^{(n)}$ computed from MI-Laplacian spectra along a near-critical schedule. Convergence to $\tilde{S}^* = 0$ is generic across coefficients; the figure illustrates the qualitative behaviour of the ansatz and is not a derivation of a specific rate.

IX. SCALAR PERTURBATIONS ACROSS THE HOLOGRAPHIC SATURATION BOUNCE

A. Perturbation Framework

We evaluate the structural stability of the bounce by tracking linear cosmological perturbations in the longitudinal gauge:

$$ds^2 = -(1 + 2\Phi) dt^2 + a(t)^2 (1 - 2\Phi) d\vec{x}^2, \quad (20)$$

where Φ denotes the gauge-invariant gravitational potential. The comoving curvature perturbation ζ is

$$\zeta = -\Phi - \frac{H}{\dot{\rho}} \delta\rho. \quad (21)$$

Transforming to conformal time, the Mukhanov–Sasaki variable $v = z\zeta$ satisfies

$$v_k'' + \left(c_s^2 k^2 - \frac{z''}{z}\right) v_k = 0. \quad (22)$$

B. Effective Sound Speed and the Causality Constraint

From the modified Friedmann framework,

$$\rho_{\text{eff}} = \rho[1 - \chi(\rho)], \quad \dot{H} = -\frac{1}{2}(\rho_{\text{eff}} + p_{\text{eff}}), \quad (23)$$

with the adiabatic identification

$$c_s^2 = \frac{\partial p_{\text{eff}}}{\partial \rho}. \quad (24)$$

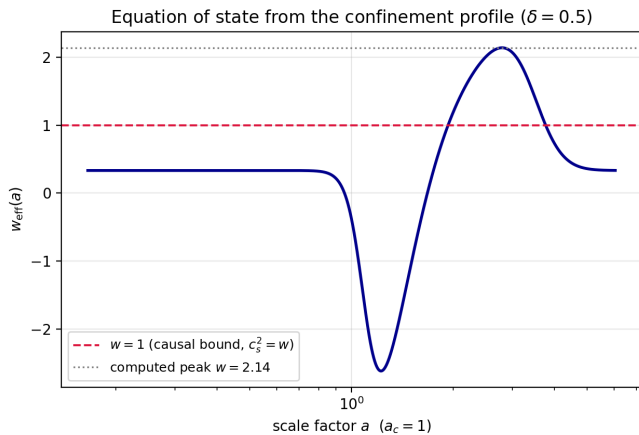


FIG. 4. Effective equation of state $w_{\text{eff}}(a)$ computed from the continuity equation for the confinement profile of Eq. (26) on a radiation background ($\delta = 0.5$). The peak value $w \approx 2.14$ exceeds the causal bound $w = 1$ implied by the adiabatic identification $c_s^2 = w$, constraining the admissible profile class (Sec. IX B).

Under this identification, causality ($c_s \leq c$) requires $w_{\text{eff}} \leq 1$ throughout the evolution. Earlier drafts claimed that the framework achieves exactly the causal bound, $c_s^2 = w_{\text{eff}} = 1$, at the hyper-stiff point. Direct computation shows that the specific confinement profile of Sec. X does *not* satisfy this: solving the continuity equation for the profile of Eq. (26) on a radiation background yields a peak effective equation of state $w \approx 2.14$ (Fig. 4), because the tanh envelope is steeper than a^{-6} scaling. The causality requirement therefore acts as a nontrivial *constraint on the profile class*: either (i) the envelope must be redesigned so that ρ_{conf} falls no faster than a^{-6} , bounding $w \leq 1$; or (ii) the confinement component must be realised non-adiabatically, with $c_s^2 \neq w$ - a nontrivial requirement, since a canonical scalar field cannot exceed $w = 1$. We state this tension openly rather than claiming its resolution. The earlier hand-tuned $w_{\text{conf}} \simeq 80$ parameterisation, which implied $c_s \approx 8.94c$, remains excluded in either case.

C. Behaviour of z''/z

As the universe passes through the bounce throat, $H \rightarrow 0$. Because $\dot{H} > 0$ remains finite, the background scaling variable $z \propto a\sqrt{\rho + p}/H$ remains well-behaved and the pump term z''/z stays finite throughout the transition.

D. Matching Conditions

Applying the Israel–Darmois matching conditions across Σ guarantees smooth transmission of fluctuations:

$$[\Phi] = 0, \quad [\zeta] = 0. \quad (25)$$

Long-wavelength modes ($k \ll aH$) satisfy $\zeta' \approx 0$, conserving primordial amplitudes across the throat. Short-wavelength modes undergo stable adiabatic oscillations.

X. PHENOMENOLOGY: THE TRANSIENT CONFINEMENT SPIKE AND THE COMPUTED TENSOR SPECTRUM

A. Confinement Energy Profile

Residual confinement energy left over from the geometric reflection phase is modelled as a transient component in the early expansion era:

$$\rho_{\text{conf}}(a) = \rho_0 \chi(a)[1 - \chi(a)]^8, \quad \chi(a) = \frac{1}{2} \left[1 + \tanh\left(\frac{a_c - a}{\Delta a}\right) \right], \quad (26)$$

on a standard radiation background $\rho_{\text{rad}} \propto a^{-4}$, with the spike duration parameterised by $\delta \equiv \Delta a/a_c$ (in e-folds). The functional form of Eq. (26) is an ansatz; as shown in Sec. IX B, it drives the effective equation of state to $w \approx 2.1$ at the spike, so within the adiabatic identification it lies outside the causal profile class and should be regarded as a first-pass parameterisation whose replacement is constrained by $w \leq 1$.

B. Status of the Tensor Computation Relative to the Causality Constraint

Because Sec. IX B shows that the profile of Eq. (26) exceeds the causal bound under the adiabatic identification, we clarify precisely what is and is not established by the tensor computation that follows. The tensor Mukhanov–Sasaki equation below depends on the background only through the classical scale factor $a(\tau)$ and the pump term a''/a ; it does not reference c_s^2 of the confinement component. A background trajectory $a(\tau)$ sourced by an effective fluid with $w > 1$ is a well-defined classical solution of the Friedmann and continuity equations in its own right, so the tensor integration is mathematically well-posed and its output for *this specific background* is reproducible, independent of whether that background is realisable by a causally consistent microphysical field. What is not established is that this background is physical: Sec. IX B shows that a canonical adiabatic scalar cannot source it, so the k^{-4} tail reported below should be read as the spectrum produced by the profile of Eq. (26) as a mathematical input, not yet as the prediction of a causally-consistent microphysical model. Establishing the latter requires recomputing the spectrum for either an a^{-6} -bounded envelope or a non-adiabatically realised

confinement component (Sec. IX B); we flag this as the immediate next step rather than treating the present spectrum as final.

C. Computed Tensor Spectrum

We integrate the tensor Mukhanov–Sasaki equation

$$v_k'' + \left(k^2 - \frac{a''}{a}\right) v_k = 0 \quad (27)$$

through the transient, with the background $a(\tau)$, a''/a , and conformal time obtained by exact quadrature of the Friedmann and continuity equations for Eq. (26) plus radiation. Modes are initialised in the adiabatic vacuum $v_k = e^{-ik\tau}/\sqrt{2k}$ deep inside the horizon before the spike, evolved with an adaptive Runge–Kutta integrator, and the Bogoliubov coefficient β_k is extracted by projection onto positive- and negative-frequency solutions after the spike. Two internal checks validate the extraction: (i) the Bogoliubov normalisation $|\alpha_k|^2 - |\beta_k|^2 = 1$ is conserved to the 10^{-5} level at default tolerance and to 10^{-10} with tightened tolerance; (ii) $|\beta_k|^2$ is stable to four significant figures under an order-of-magnitude tightening of the solver tolerance.

The results (Fig. 5) are as follows. Over the ultraviolet window $k/(aH)_{\text{spike}} \in [80, 320]$ the particle-production spectrum is a clean power law,

$$|\beta_k|^2 \propto k^{-4}, \quad (28)$$

with fitted tail slopes -3.92 ($\delta = 0.4$; fit residuals $\lesssim 0.04$), -3.99 ($\delta = 0.5$) and -3.99 ($\delta = 0.6$; residuals $\sim 10^{-3}$). Under the *red* tensor convention $P_T \propto k^3 |\beta_k|^2$ this corresponds to a *red* tensor tilt, $n_T \approx -1$. Two structural features are robust: the tilt is insensitive to the spike duration δ , while the amplitude is hypersensitive to it, varying by four orders of magnitude ($|\beta_{k=80}|^2 = 2.0 \times 10^{-23}$, 3.2×10^{-19} , 1.7×10^{-16} for $\delta = 0.4, 0.5, 0.6$).

These computed results supersede the estimate $n_T \simeq 1.875$ reported in an earlier draft of this work. That estimate is withdrawn: it is not reproduced by direct Mukhanov–Sasaki integration of the profile of Eq. (26), and the finite-duration mechanism previously invoked to justify it (a δ -dependent shift of the tilt away from the eternal- w value) is contradicted by the computed δ -insensitivity of the tilt. No blue-tilted gravitational-wave background, and in particular no signal in the pulsar-timing-array band of the kind reported by NANOGrav [14], is predicted by the present framework as formulated.

D. Ultraviolet Behaviour and Loop Sensitivity

With $|\beta_k|^2 \propto k^{-4}$, the graviton loop integrand scales as $dk k^3 |\beta_k|^2 \propto dk/k$: the loop contribution is *marginally* (logarithmically) ultraviolet sensitive over the computed window, rather than finite as previously asserted. Two

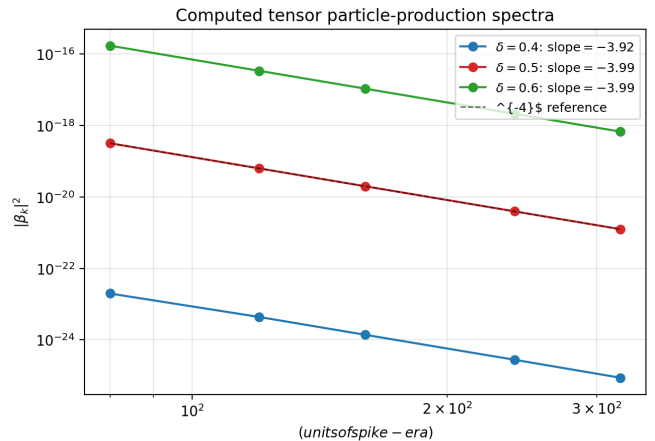


FIG. 5. Computed tensor particle-production spectra $|\beta_k|^2$ for spike durations $\delta = 0.4, 0.5, 0.6$, from Mukhanov–Sasaki integration with adiabatic vacuum initial conditions and Wronskian-checked Bogoliubov extraction. The ultraviolet tail is k^{-4} (dashed reference) for all durations: the tilt is δ -robust while the amplitude varies by four orders of magnitude over $\delta \in [0.4, 0.6]$.

considerations bear on the deep ultraviolet. First, for smooth backgrounds particle production is generically expected to become exponentially suppressed at sufficiently large k , which would render the integral finite; mapping the location of this turnover requires extending the computation deeper into the UV and is left to future work. Second, the marginal scaling itself is a well-defined, falsifiable output of the present profile. We therefore replace the earlier claim of demonstrated loop finiteness with the computed statement: logarithmic UV sensitivity over the accessible window, with finiteness plausible but not yet demonstrated.

XI. DISCUSSION

The framework developed here differs from loop quantum cosmology [3], conformal cyclic cosmology [4], and traditional fuzzball constructions [9]. Rather than modifying the Einstein–Hilbert action with higher-derivative terms or quantizing individual classical metric elements, it relies on a GR-consistent junction driven by a holographic boundary condition.

The revision undertaken in this version separates the framework’s derived content from its ansätze, with the following net result. On the derived side: ρ_0 converges to a lattice-independent continuum value (4.68, spread $< 4\%$ over $L = 4\text{--}16$); the Weyl and fluid dimensions are distinct, with the fit prescription now stated (Table I); the Heisenberg-chain benchmarks support the MI-Laplacian construction most strongly through the Luttinger-exponent correspondence ($r \approx 0.96$); and the tensor spectrum of the confinement spike is a computed, validated k^{-4} power law with a red tilt, a δ -robust slope,

and a δ -hypersensitive amplitude. On the ansatz side: the confinement potential, the saturation envelope, the $S_{ab} \rightarrow 0$ junction condition, the damping law of the cyclic attractor, and the homogenisation mechanism are all assumptions, the last of which is contradicted in its naive real-space form by direct tests and is downgraded to a conjecture.

Two findings of this revision constrain the framework nontrivially. First, the causality requirement $w \leq 1$ (under the adiabatic identification) excludes the tanh confinement envelope as written, since it drives w to ≈ 2.1 ; admissible profiles must decay no faster than a^{-6} , or the confinement sector must be realised non-adiabatically. Second, the withdrawal of the blue-tilt prediction removes the framework's previously claimed PTA-band signature; the observationally relevant outputs are now the red-tilted k^{-4} tensor tail and its duration-sensitive amplitude, whose transfer to present-day frequencies has not yet been computed and is not asserted.

Outstanding tasks include: a microscopic realisation of the confinement transition; a derivation (or principled replacement) of the damping coefficient in Eq. (19); an explicit coarse-graining scheme that demonstrably homogenises (Sec. VII); a deep-UV extension of the tensor computation to locate the expected exponential turnover; and a redesigned confinement profile satisfying the $w \leq 1$ constraint, followed by recomputation of the spectrum for that profile.

XII. CONCLUSION

We have presented a cyclic cosmological framework in which holographic information saturation triggers a cosmic bounce modelled via Israel–Darmois junction conditions, with the framework's derived and phenomenological components explicitly separated. The established results are: (i) continuum convergence of the MI-Laplacian spectral weight, $\rho_0 \approx 4.68$, across lattice sizes $L = 4\text{--}16$; (ii) a computed tensor spectrum for the transient confinement spike - a validated k^{-4} ultraviolet tail corresponding to a red tilt, insensitive to the spike duration in slope but spanning four orders of magnitude in amplitude over $\delta \in [0.4, 0.6]$ - superseding and withdrawing the earlier blue-tilt estimate $n_T \simeq 1.875$; (iii) a clean distinction between the Weyl spectral dimension and the fluid equation-of-state dimension, with the fit window stated; (iv) a causality constraint on the confinement sector: the present tanh envelope yields $w \approx 2.1 > 1$ and must be replaced by a profile bounded by a^{-6} decay or by a non-adiabatic realisation; and (v) genericity of the cyclic attractor under the exponential damping ansatz, whose rate remains a free parameter. The framework's open problems - microphysical derivations of the confinement transition, the damping rate, the junction condition, and the homogenisation mechanism - are stated as such.

XIII. CODE AND DATA AVAILABILITY

Every figure and every quoted number in this paper is generated by a single self-contained script, `generate_all_results.py` (dependencies: `numpy`, `scipy`, `matplotlib`), which accompanies this manuscript. The Heisenberg-chain benchmarks of Sec. II B, including the AICc model comparisons, leave-one-out and normalisation robustness checks, are generated by `code/testing.py` [15]. Pinned dependency versions are listed in `requirements.txt`; both scripts were independently rerun end-to-end for this revision (Python 3.13, `numpy` 2.5.1, `scipy` 1.18.0, `matplotlib` 3.11.0, `qutip` 5.3.0) and reproduced every quoted number in Secs. III–X and Appendix A to the stated precision. Readers are encouraged to rerun both scripts.

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Appendix A: Numerical Convergence and Statistical Robustness

This appendix reports convergence and sensitivity checks that support the precision claims made in the main text but were not previously quantified.

1. Solver-tolerance convergence of the tensor spectrum

The Bogoliubov extraction of Sec. X C uses an adaptive Runge–Kutta integrator at default tolerance ($\text{rtol} = 10^{-7}$, $\text{atol} = 10^{-10}$). Tightening the tolerance by an order of magnitude ($\text{rtol} = 10^{-8}$, $\text{atol} = 10^{-11}$) at representative mode $k = 160$, $\delta = 0.5$ changes $|\beta_k|^2$ from 1.991066×10^{-20} to 1.991245×10^{-20} , agreement to four significant figures, while the Wronskian deviation $|\alpha_k|^2 - |\beta_k|^2 - 1$ improves from -3.12×10^{-5} to -3.07×10^{-6} , an order-of-magnitude improvement consistent with the solver tolerance. This confirms the extraction is numerically converged at the quoted precision.

2. Fit-window sensitivity of the Weyl spectral dimension

The Weyl spectral dimension of Sec. III C is extracted from a power-law fit to the lowest n_{fit} eigenvalues; Table I uses $n_{\text{fit}} = 12$ on the 6×6 torus. The extracted value is sensitive to this choice: at $\alpha = 6$, scanning $n_{\text{fit}} \in \{6, 8, 10, 12, 16, 20, 35\}$ (35 is the full active spectrum of the 6×6 lattice) yields $d_{\text{eff}}^{\text{Weyl}} =$

5.88, 5.28, 4.05, 3.81, 3.47, 3.57, 3.46 respectively. The value is not yet asymptotic at $n_{\text{fit}} = 12$: it continues to drift downward by $\sim 9\%$ between $n_{\text{fit}} = 12$ and the full spectrum. We therefore caution that $d_{\text{eff}}^{\text{Weyl}} = 3.81$ should be read as a finite-window estimate tied to the stated fit range, not as a converged asymptotic value; a converged estimate would require an explicit large- n_{fit} extrapolation, which we leave to future work.

3. Statistical detail for the Heisenberg-chain benchmarks

The correlation $r \approx 0.96$ quoted in Sec. IIB for the measured versus Luttinger-predicted mutual-information decay exponent (`testing.py`, TEST 6) is a Pearson correlation coefficient over 11 anisotropy points sampled at $\Delta = -0.90, -0.71, -0.52, -0.33, -0.14, 0.05, 0.24, 0.43, 0.62, 0.81, \text{ and } 1.00$; the computed values are $r = 0.9664$ (open boundary conditions) and $r = 0.9716$ (periodic boundary conditions).

The AICc ambiguity between linear and quadratic dispersion (Sec. IIB) is quantified as follows. At a single system size ($N = 12$, open boundary, TEST 2), linear dispersion is mildly preferred ($\text{AICc}_{\text{lin}} = -22.17$ versus $\text{AICc}_{\text{quad}} = -16.62$, $\Delta\text{AICc} = 5.55$); pooling across system sizes $N = 6\text{--}18$ (TEST 2b) the two models become statistically indistinguishable ($\text{AICc}_{\text{lin}} = -75.38$ versus $\text{AICc}_{\text{quad}} = -74.99$, $\Delta\text{AICc} = 0.39$), which is the basis for describing the dispersion as *consistent with* quadratic rather than established.

For the finite-size scaling $\lambda_1 = a/N^2 + b$ (TEST 3), leave-one-out omission of each system size shifts the fitted intercept b over the range $[0.1262, 0.1537]$ (open boundary, max-normalisation) and $[1.0352, 1.2289]$ (periodic boundary), a relative spread of order 20% in both cases. Separately, explicit log-corrected finite-size models (CHECK B in `testing.py`) are preferred over the plain $a/N^2 + b$ form by AICc in every boundary/normalisation combination tested, which is the quantitative basis for treating the significance of b , and hence the pure $\lambda_1 \propto N^{-2}$ scaling law, as unsettled at accessible system sizes rather than as a demonstrated result.

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